

Yvonne Zhang

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in Yvonne Zhang

📁 Portfolio

Education

Johns Hopkins University

Master's of Science in Engineering in Robotics

Baltimore, MD

August 2025 – June 2027

University of Toronto

Bachelor's of Applied Science in Mechanical Engineering & PEY Co-op

Toronto, Canada

September 2020 – June 2025

Project Experience

Autonomous Bleeding Control in Partial Nephrectomy

Baltimore, MD

IMERSE Lab, Johns Hopkins University

February 2026 – May 2026

- Fabricated patient-specific hydrogel kidney phantoms with silicone perfusion channels to simulate controlled intraoperative bleeding for robot-assisted partial nephrectomy.
- Developed a YOLO and U-Net++ vision pipeline for bleeding localization and segmentation using annotated dVRK endoscopic images and videos from 3D Slicer.
- Extending the platform to real-time bleeding source localization and classification for autonomous dVRK robotic control of suction and cautery assistance.

Visual Information through Sensory Tactile Array (VISTA)

Baltimore, MD

HAMR Lab, Johns Hopkins University

August 2025 – Present

- Designed a 7×5 tactile pin-array display to translate visual shape and color into height-encoded tactile representations for accessible museum exhibits.
- Developed a Python image-processing pipeline and Arduino closed-loop controller to actuate 35 motors for real-time tactile rendering.
- Evaluated the system through user studies on tactile recognition of flag shapes and colors; presented a Work-in-Progress paper and poster at the IEEE Haptics Symposium 2026.

UR5 Manipulator Foundations

Baltimore, MD

Johns Hopkins University

February 2026 – May 2026

- Developed POE-based forward kinematics, Jacobian motion control, and Cartesian trajectory generation for a UR5 using C++ and ROS2.
- Implemented hand-eye calibration and probabilistic roadmap motion planning with collision checking and A* search for autonomous manipulation.
- Validated full manipulation pipeline on a physical UR5 following simulation-based testing in RViz and MoveIt!.

Gesture-Based Surgical Robot Assistance

Zurich, Switzerland

pd/z Product Development Group Zurich, ETH Zurich

June 2024 – September 2024

- Designed a dual Intel RealSense D435 end-effector mount for a Franka Panda, defining reference frames for intraoperative depth sensing.
- Developed ROS2 perception pipelines for real-time depth extraction and 12-class hand gesture recognition using MediaPipe and custom keypoint classifiers.

Window Cable Tensioner Redesign

Toronto, Canada

Magna International & University of Toronto

September 2023 – December 2023

- Redesigned a frameless car-door cable tensioner by identifying failure modes of fatigue, wear and stress concentrations in plastic components and spring mechanism, while preserving one-way cable tensioning functionality.
- Leveraged SolidWorks CAD and Simulation to iterate four design variants, validating structural performance and reducing manufacturing cost.

Work Experience

Systems Engineer Internship

Toronto, Canada

WSP Canada

May 2023 – May 2024

- Authored a novel methodology for characterizing thermal properties of train fire spread using material testing data, informing CFD boundary conditions and adopted across the Calgary Green Line project.
- Performed hydraulic modeling and fire line design in AFT Fathom 13 to ensure compliance with NFPA 130 and NFPA 14 volumetric flow and pressure constraints.

Technical Skills

Programming: Python, MATLAB, Arduino IDE, C, C++

Software & Analysis: SolidWorks, ANSYS Mechanical and Fluent